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Transfer Learning for Spatial Autoregressive Models with Application to U.S. Presidential Election Prediction

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ABSTRACT

It is important to incorporate spatial geographic information into U.S. presidential election analysis, especially for swing states. The state-level analysis also faces significant challenges of limited spatial data availability. To address the challenges of spatial dependence and small sample sizes in predicting U.S. presidential election results using spatially dependent data, we propose a novel transfer learning framework within the SAR model, called as tranSAR. Classical SAR model estimation often loses accuracy with small target data samples. Our framework enhances estimation and prediction by leveraging information from similar source data. We introduce a two-stage algorithm, consisting of a transferring stage and a debiasing stage, to estimate parameters and establish theoretical convergence rates for the estimators. Additionally, if the informative source data are unknown, we propose a transferable source detection algorithm using spatial residual bootstrap to maintain spatial dependence and derive its detection consistency. Simulation studies show our algorithm substantially improves the classical two-stage least squares estimator. We demonstrate our method's effectiveness in predicting outcomes in U.S. presidential swing states, where it outperforms traditional methods. In addition, our tranSAR model predicts that the Republican Party would win the 2024 U.S. presidential election.

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High-dimensional analysis; Spatial autoregressive model; Spatial data analysis; Transfer learning; U.S. presidential election

1. Introduction

In recent years, large amounts of Geographic Information Science (GIS) data have been used in election campaigns (Peters, DeYoung, and Ross 2004). Geography is the common denominator of where voters live, how residents engage in the voting process, how officials manage elections, and where election campaigns run. Based on population data and geographic information, election district boundaries are resettled to draft multiple proposed plans and demonstrate the effects of several approaches. Political campaigns make full use of geographic information and related tools to obtain underlying support from voters, especially the ones from swing states, which are pivotal in determining the outcome of elections. Swing states, also known as battleground states, are crucial because they do not consistently support one party, making their election outcomes unpredictable and highly influential in election results. Shaping voter preference in swing states is an essential problem, especially for prediction. Including spatial information in data analysis brings a novel insight into the prediction. There are at least two challenges in using spatial information for predicting U.S. presidential elections: one is leveraging spatial information effectively to predict election outcomes while considering the impact of spatial factors, and the other is the limited data from swing states. Finding ways to use information from other states to analyze spatial factors and predict the final result is crucial for overcoming these challenges.

Spatial dependence exists widely in various fields including economics (Anselin 1988), finance (Grennan 2019), political science (Peters, DeYoung, and Ross 2004), ecology (Ver Hoef et al. 2018), and climate sciences (Okunlola et al. 2021). The spatial autoregressive (SAR) model (Cliff and Ord 1975) and its variants are extensively applied in numerous empirical studies concerning spatial competition and spillover effects. For instance, the SAR model is employed to investigate the crime rates in 49 areas of Columbus, Ohio (Anselin 1988), budget spillover effects and policy interdependence (Case, Rosen, and Hines 1993), cigarette demands in the U.S. (Baltagi and Li 2004), agricultural land prices in Northern Ireland (Kostov 2009), and the mutual influence on the stock returns of companies with the same headquarters location (Pirinsky and Wang 2006).

When the sample size of the target data is limited, it is possible to obtain less precise estimates for most statistical models. This issue is particularly relevant in spatial econometrics, where sample sizes can often be quite limited. For instance, in the study investigating crime rates in Columbus, Ohio, the sample size is restricted to just 49 (Anselin 1988). This limitation underscores the necessity for additional research aimed at leveraging information from similar datasets to elevate estimation performance. Consequently, there is a pressing need for the exploration of methods to improve the estimation performance of the SAR

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models through the incorporation of information derived from similar source datasets. Transfer learning is a useful approach to incorporate information from source datasets (Hu and Zhang 2023; Pan and Yang 2010; Torrey and Shavlik 2010). The main idea of transfer learning is that the knowledge acquired in one context can be transferred to improve the performance of estimation or prediction when applied to another, even if the two tasks are not identical. In transfer learning, there are typically two main components: (1) The first one is the *target domain*, to which the knowledge is transferred. The target domain is of our interest, but its sample size is particularly limited. (2) The second one is the *source domain*, from which the knowledge is transferred. In general, the model is pre-trained using the source datasets.

In the recent literature, Bastani (2021) introduced a two-step estimator that employs regularization techniques to capture biases between the target and source data and has provided an upper bound for the proposed estimators. Li, Cai, and Li (2022b) considered transfer learning for estimation and prediction of a high dimensional linear regression and proposed a novel data-driven Trans-Lasso method for detecting the transferable sources. Tian and Feng (2023) introduced a novel entropy-based method for identifying transferable sources and further constructed confidence intervals for each coefficient under the generalized linear model. For more reference on high-dimensional independent data setting, see Cai and Pu (2019), Cai, Cai, and Li (2024a), Cai, Kim, and Pu (2024b), Bai et al. (2025), Chen et al. (2024), Wang et al. (2025), Tian, Gu, and Feng (2025), Tripuraneni, Jin, and Jordan (2021) and Li, Cai, and Li (2022a). Regarding dependent data structures, He et al. (2023), and Ma and Safikhani (2025) explored transfer learning to time series analysis, while Ren, Zhen, and Wang (2024) studied transfer learning for tensor Gaussian graphical models. However, the literature on transfer learning specifically for spatial data remains unexplored, which is the main focus of this paper.

In this paper, in order to deal with two aforementioned challenges, spatial dependence and sample limitation, in U.S. presidential election prediction problems, we present a transfer learning framework within the SAR model, which we refer to as **tranSAR**, designed to improve both estimation and prediction. Our study makes three contributions from methodology, statistical theory and empirical application in the literature of the SAR models. First, in methodology, we introduce a two-stage algorithm, $\mathcal{A}$-TranSAR, consisting of a transferring stage and a debiasing stage, to estimate the unknown parameters for a user-specified set $\mathcal{A}$. If we do not know which sources to positively transfer, a transferable source detection algorithm is proposed to detect informative sources data based on spatial residual bootstrap to retain the spatial dependence. To the best of our knowledge, this is the first work to introduce the transfer learning to the SAR models. Second, in statistical theory, we establish the theoretical convergence rates for the $\mathcal{A}$-TranSAR estimators under the SAR model and theoretically show that the transfer learning can improve the estimation accuracy with the help of the information from the transferable source data. We also derive the detection consistency of the transferable source detection algorithm to avoid the negative transfer when which sources to transfer are unknown. Third, in the empirical application, we apply our method to predict the election outcomes in swing states in the U.S. presidential election via utilizing polling data from the last U.S. presidential election along with other demographic and geographical data. The empirical results show that our method outperforms traditional spatial econometrics methods.

Here, we emphasize the differences between our work and Tian and Feng (2023). Although our work shares a similar idea of transfer leaning with the previous literature, there are at least two significant differences from both theoretical and algorithmic perspectives. (1) Tian and Feng (2023) is based on generalized linear models for independent data. In contrast, our research focuses on spatially dependent data, which brings significant challenges for theoretical guarantees. This is because previous theories heavily rely on concentration inequalities applicable to independent data, and few concentration inequalities have been established for spatially dependent data. To address this, we developed a new theoretical foundation tailored to spatially correlated data. (2) For the algorithmic aspect, the transferable detection method in Tian and Feng (2023) relies on sample splitting to identify transferable sources, a technique that is inapplicable for spatial data as spatial data are not independent. To this end, we propose a new method based on spatial bootstrap and establish its theoretical guarantees.

In the rest of our paper, we firstly introduce the real application problem (predicting the U.S. presidential election result) and the associated transfer learning framework for the SAR model in Section 2. Then, we introduce its estimation and source detection algorithm in Section 3. Section 4 establishes the estimation convergence rates of the proposed methods for the SAR model and derives the detection consistency of the transferable source detection algorithm. Next, we evaluate the finite sample performance of our proposed method via simulation studies in Section 5. Further, the application to predict U.S. presidential elections with transfer learning is conducted in Section 6. Finally, we conclude the paper in Section 7. Additional simulation results, theoretical analysis, and all proofs are provided in the supplementary materials, and the code is available at the anonymous repository https://anonymous.4open.science/r/tranSAR-E508.

Notations. Let $[n] \equiv \{1, \ldots, n\}$. For an index set $\mathcal{A} \subseteq [K]$ and a sequence of positive integers $\{n_k, k = 1, \ldots, K\}$, $n_{\mathcal{A}} \equiv \sum_{k\in\mathcal{A}} n_k$. For a vector $\mathbf{x} \in \mathbb{R}^d$ and S being a subset of set $[d]$, $\mathbf{x}_S$ is the sub-vector of $\mathbf{x}$ with index S. The notation $a \lesssim b$ means there exists a positive constant C s.t. $a \leq Cb$. The notation $a \vee b$ means $\max(a, b)$, and $a \wedge b$ means $\min(a, b)$. The notation $a_n \asymp b_n$ means that a_n/b_n converges to a positive constant. For any random sequences a_n and b_n, $b_n = O_{\mathbb{P}}(a_n)$ means $\forall \epsilon > 0, \exists M > 0$ s.t. $\limsup_n \mathbb{P}(|b_n| \geq M|a_n|) \leq \epsilon$, and $a_n \ll b_n$ means $|a_n| = o_{\mathbb{P}}(|b_n|)$, i.e. $\lim_{n\to\infty} \mathbb{P}(|\frac{a_n}{b_n}| \geq \varepsilon) = 0$ for any $\varepsilon > 0$. For a vector $\mathbf{x} = (x_1, \ldots, x_d)^\intercal \in \mathbb{R}^d$, we define $\| \mathbf{x} \|_1 = \sum_{j=1}^d |x_j|$, $\| \mathbf{x} \|_2 = (\sum_{j=1}^d x_j^2)^{1/2}$, and $\| \mathbf{x} \|_\infty = \max_{1\leq j\leq d} |x_j|$. For vectors $\mathbf{x}$ and $\mathbf{y} \in \mathbb{R}^d$, their inner product is $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^\intercal \mathbf{y}$. For an $n \times n$ matrix A, we define $\| A \|_F \equiv (\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2)^{1/2}$, $\| A \|_\infty \equiv \max_i \sum_{j=1}^n |a_{ij}|$, its spectral norm $\| A \|_2 \equiv \max_{x\in\mathbb{R}^n\setminus\{0\}} \frac{\|Ax\|_2}{\|x\|_2} = \sqrt{\lambda_{\max}(A'A)}$, where $\lambda_{\max}(A)$ is the maximum eigenvalue of A and $\lambda_{\min}(A)$ is the minimum

eigenvalue of A. For an $n \times m$ matrix A, $A_{(i,j)}$ represents the element in its i-th row and j-th column. $A_{(\cdot,j)}$ refers to the j-th column of A, and $A_{(i,\cdot)}$ represents the i-th row of matrix A.

2. Problem and Model Settings

In this work, we focus on predicting the U.S. presidential election results of each county in swing states by incorporating the spatial relationship among counties. We try to predict the election results in 2020 and 2024 elections using the polls from the 2016 and 2020 election U.S. presidential election and some other demographic data and geographic data, respectively. It is worth noting that swing states have undergone historical shifts over time. The process of identifying swing states in previous elections typically involves an assessment of the closeness of the vote margins in each state. This analysis incorporates historical election outcomes, opinion polls, political trends, recent developments since the previous election, and the specific attributes, strengths, or vulnerabilities of the candidates in contention. In Section 2.1, we describe the motivation problem and data. Here, we reiterate the challenges we face, namely the utilization of spatial information and data scarcity. To this end, we propose a novel transfer learning framework based on the SAR model aimed at addressing above-mentioned two challenges.

2.1. Problem and Data

2.1.1. Problem

We aim to handle the challenges of utilizing spatial information and analyzing spatial factors in predicting the U.S. presidential election. From Figure 1, it is evident that the county-level support rates in the U.S. presidential election display a spatial relationship. Spatial relationships among the data are also prevalent, making it essential to incorporate spatial information into the analysis. In U.S. presidential elections, the distribution of votes exhibits significant spatial dependency, a phenomenon particularly obvious over counties. Specifically, the voting patterns of a county tend to align closely with those of its neighboring counties. This similarity can often be attributed to the shared socio-economic characteristics that arise from geographical proximity, such as industrial structure, income levels, and educational attainment (Fotheringham, Li, and Wolf 2021; Kim, Elliott, and Wang 2003). Additionally, whether a county is urban or rural plays a substantial role in this spatial dependency. Urban counties generally show stronger support for the Democratic Party, whereas suburban and rural counties tend to favor the Republican Party. Thus, in analyzing voting distributions, the urban or rural status of a county becomes a critical factor, reflecting not only the geographical distribution of political preferences but also highlighting the political and social divides between urban and rural areas. Recognizing this spatial dependency helps in more accurately predicting election outcomes and in forming strategies for political parties. Thus, the first challenge lies in how to incorporate spatial information into transfer learning techniques. Though the method proposed by Tian and Feng (2023) provides a valuable framework to transfer learning and has been applied to the U.S. presidential election, their method is for independent data and does not account for spatial information during transfer learning. Furthermore, the second challenge is the issue of data scarcity. It is evident that the average sample size per state (the number of counties) is only around 60. Relying only on data from swing states for spatial analysis could harm the accuracy of the estimates.

2.1.2. Data Sources and Processing

The digital boundary definitions of the United States congressional district are available at https://cdmaps.polisci.ucla.edu/, and the cartographic boundary shape files of counties in the USA are available at the United States Census Bureau (https://www2.census.gov/geo/tiger/). The spatial weight matrix is constructed using above two geographic datasets. We apply the "queen" contiguity-based spatial weights and row-normalized. The "queen" criterion defines neighbors as spatial units sharing a common edge or a common vertex. The response is the difference of support at the county level from both parties in 2016 and 2020, available at https://github.com/tonmcg/US_County_Level_Election_Results_08-20. The county-level demographic information forms the predictors, collected at https://data.census.gov/. Among all states and the federal district, we exclude Alaska, Hawaii, and Washington, D.C. In these 48 states, there are 3107 counties with 142 county-level predictors.

2.2. Model Settings

In order to deal with the two aforementioned challenges in U.S. presidential election prediction problems, we present a transfer learning framework within the spatial autoregressive (SAR) model. For a pre-specified swing state as the target data, we consider the following SAR model,

$$Y^{(0)} = \sum_{l=1}^{p} \lambda_{l0} W_l^{(0)} Y^{(0)} + \sum_{j=1}^{q} X_j^{(0)} \beta_{j0} + V^{(0)}, \tag{1}$$

where the difference of support $Y^{(0)}$ is an $n_0 \times 1$ spatially correlated vector, $W_l^{(0)}$ is an $n_0 \times n_0$ known nonstochastic spatial weight matrix for each $l \in [p]$, the county-level demographic information $X_j^{(0)}$ is an $n_0 \times 1$ vector of exogenous regressors independent of the error term $V^{(0)}$ for each $j \in [q]$, $V^{(0)}$ is an $n_0 \times 1$ vector of independently and identically distributed (i.i.d.) random variables with mean 0 and variance $\sigma^2 > 0$. Let $\lambda_0 = (\lambda_{10}, \ldots, \lambda_{p0})^\intercal$, $\beta_0 = (\beta_{10}, \ldots, \beta_{q0})^\intercal$ and $\theta_0 \equiv (\lambda_0{}^\intercal, \beta_0{}^\intercal)^\intercal$.

Similar to general transfer learning settings as in Bastani (2021), Li, Cai, and Duan (2023), and Tian and Feng (2023), in addition to the target study (1), additional samples are obtained from K auxiliary studies (also termed as *source datasets*) from other states, where K is assumed to be fixed. For the k-th source dataset, we also consider the SAR model,

$$Y^{(k)} = \sum_{l=1}^{p} \lambda_{l0}^{(k)} W_l^{(k)} Y^{(k)} + \sum_{j=1}^{q} X_j^{(k)} \beta_{j0}^{(k)} + V^{(k)}, \quad k \in [K], \tag{2}$$

where $Z^{(k)} \equiv (Y^{(k)}, X_1^{(k)}, \ldots, X_q^{(k)})$ is the k-th source dataset of sample size n_k, $W_l^{(k)}$ is an $n_k \times n_k$ nonstochastic spatial weight matrix for each $l \in [p]$, and $V^{(k)}$ is an $n_k \times 1$ vector of i.i.d. random variables with mean zero and variance σ_k^2, independent of the covariates $X_j^{(k)}, j \in [q]$. Let $\lambda_0^{(k)} = (\lambda_{10}^{(k)}, \ldots, \lambda_{p0}^{(k)})^\intercal$,

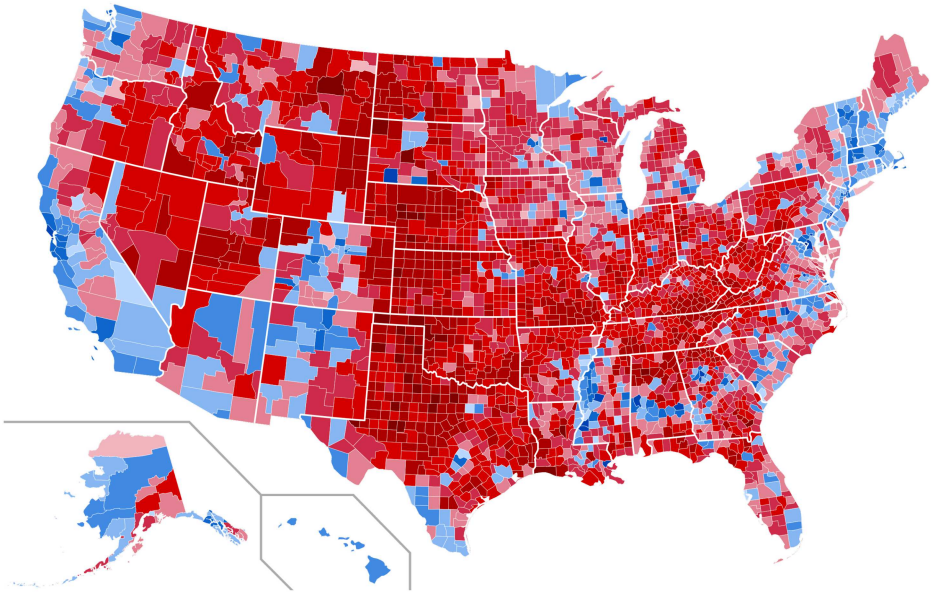


Figure 1. Results by county of the 2020 United States presidential election (Magog the Ogre 2020). The red areas represent support for Trump, while the blue areas indicate support for Biden. The intensity of the color corresponds to the degree of support.

$\beta_0^{(k)} = (\beta_{10}^{(k)}, \ldots, \beta_{q0}^{(k)})^\intercal$ and $\omega_0^{(k)} \equiv (\lambda_0^{(k)\intercal}, \beta_0^{(k)\intercal})^\intercal$ for the k-th source dataset. Let $\mathbf{X}^{(k)} \equiv (X_1^{(k)}, \ldots, X_q^{(k)})$, $\mathbf{W}^{(k)} \equiv (W_1{}^{(k)}, \ldots, W_p{}^{(k)})$, and $\mathfrak{D}^{(k)} \equiv (Y^{(k)}, \mathbf{X}^{(k)}, \mathbf{W}^{(k)})$, $k = 0, \ldots, K$. In transfer learning, an auxiliary model similar to the target model is called an *informative* model, and the "similarity" is characterized by the difference between $\omega_0^{(k)}$ and θ_0, i.e. $\delta^{(k)} \equiv \theta_0 - \omega_0^{(k)}$. Given a tolerance level $h > 0$, the index set of the informative auxiliary samples is defined as

$$\mathcal{A} \equiv \mathcal{A}_h = \{0 \le k \le K : \| \delta^{(k)} \|_1 \le h\}, \tag{3}$$

where $\| \delta^{(k)} \|_1$ is called the *transferring level* of the k-th source dataset.

Remark 1. In one hand, we could image that there is an oracle tolerance level h determining the informative set $\mathcal{A}$ beforehand. In this context, the value of h is used to balance the bias and the variance introduced by transfer learning. When h is large, more labeled datasets are used, which reduces the prediction variance but increases the bias due to source datasets, and vice versa. In another hand, if user specifies the informative set $\mathcal{A}$ beforehand, we could also define h as the maximum transferring level of the source datasets in the informative set, that is,

$$h \equiv h_{\mathcal{A}} = \max_{k \in \mathcal{A}} \| \delta^{(k)} \|_1 . \tag{4}$$

When user specifies the set $\mathcal{A}$, the larger the set $\mathcal{A}$, the larger the h. It means the bias-variance tradeoff similarly. The bias-variance tradeoff is a key consideration in the transfer learning literature: with an unsuitable h or user-specified set $\mathcal{A}$, the transfer learning may introduce too much bias or too much variance and harm the prediction accuracy. It is also called as negative transfer (Tian and Feng 2023). We will numerically show the impact of negative transfer in Section 5. To avoid the negative transfer, we propose a transferable source detection algorithm to detect the informative set $\mathcal{A}$ automatically in Section 3.2 after the method $\mathcal{A}$-TranSAR in Section 3.1 for a given informative set $\mathcal{A}$. We will discuss h in details about theoretical requirements in Section 4.

3. Statistical Methodology

In this section, we introduce transfer learning methods under the SAR model, including estimation with a user-specified set of source data in Section 3.1 and a transferable source detection algorithm to avoid the negative transfer in Section 3.2. Both are summarized in Algorithm 3.1 and 3.2.

3.1. Estimation with User-Specified Set of Sources $\mathcal{A}$

In this section, we propose a two-step transfer learning method under the SAR model when the index set $\mathcal{A}$ of the informative auxiliary samples is user-specified beforehand, denoted as $\mathcal{A}$ - *TranSAR*. In the first transferring step, we consolidate information from various sources by pooling all available auxiliary data, yielding a preliminary estimator which is typically biased, since $\omega_0^{(k)} \neq \theta_0$ in general. In the second debiasing step, we correct the estimation bias by incorporating the target data via regularization. The details of the two-stage estimation procedure are present as follows.

3.1.1. The Transferring Stage

The first-step estimator, which is also called *pre-training* model in the transfer learning literature, with tuning parameter λ_ω is defined as

$$\widehat{\omega} = \underset{\omega \in \mathbb{R}^{p+q}}{\arg\min} \left\{ \frac{1}{2n_{\mathcal{A}}} \sum_{k \in \mathcal{A}} \ell_1 \left(\mathfrak{D}^{(k)} \mid \omega \right) + \lambda_\omega \| \omega \|_1 \right\}, \tag{5}$$

where $\ell_1(\cdot \mid \omega)$ is a generic loss function with parameter ω, and $n_{\mathcal{A}} \equiv \sum_{k \in \mathcal{A}} n_k$ is the total sample size of all informative source data. For the SAR model, the estimator could be the two-stage least squares (2SLS) estimator (Kelejian and Prucha 1998; Lee 2003), generalized method of moments estimator (GMM) (Kelejian and Prucha 1999), quasi-maximum likelihood estimator (Lee 2004, QMLE) and so on. The main requirement for the

Algorithm 1 $\mathcal{A}$ *-TranSAR*

Input: target dataset: $\mathfrak{D}^{(0)}$; sources datasets: $(\mathfrak{D}^{(k)}, k \in \mathcal{A})$; transferring set: $\mathcal{A}$; and tuning parameters: λ_ω and λ_δ

Output: $\hat{\theta}_{\mathcal{A}\text{-}TranSAR}$

1 **The transferring stage:**

$$\widehat{\omega} = \underset{\omega \in \mathbb{R}^{p+q}}{\arg\min} \left\{ \frac{1}{2n_{\mathcal{A}}} \sum_{k \in \mathcal{A}} \ell_1 \left(\mathfrak{D}^{(k)} \mid \omega \right) + \lambda_\omega \|\omega\|_1 \right\}$$

2 **The debiasing stage:**

$$\widehat{\delta} = \underset{\delta \in \mathbb{R}^{p+q}}{\arg\min} \frac{1}{2n_0} \ell_2 \left(\mathfrak{D}^{(0)} \mid \widehat{\omega} + \delta \right) + \lambda_\delta \|\delta\|_1.$$

3 **Output:** $\widehat{\theta}_{\mathcal{A}\text{-}TranSAR} = \widehat{\omega} + \widehat{\delta}$

first step estimator is a suitable convergence rate, discussed in detail in Section 4. Here the tuning parameter λ_ω could be zero in low-dimensional cases. When the dimension of parameters is greater than the sample size in high-dimensional cases, the penalized SAR estimation with nonzero λ_ω could produce a sparse estimator (Higgins and Martellosio 2023). See also Zhang and Yu (2018) for spatial weight matrices selection with model averaging, and Tibshirani (1996), Fan and Li (2001), and Zhang (2010) for regularization method.

Remark 2. The index set $\mathcal{A}$ may or may not contain the index 0. This depends on the actual application. In some cases the source data are available, and in the other cases, it may be difficult to obtain due to data privacy protection and security considerations. In the latter case, the owner of the source data can provide the estimation results obtained by the first stage learner without the target data, and the target task will use this result for the next debiasing stage.

3.1.2. The Debiasing Stage

In the second step, the bias term estimator $\widehat{\delta}$ with tuning parameter λ_δ using the target data is defined as

$$\underset{\delta \in \mathbb{R}^{p+q}}{\arg\min} \frac{1}{2n_0} \ell_2 \left(\mathfrak{D}^{(0)} \mid \widehat{\omega} + \delta \right) + \lambda_\delta \parallel \delta \parallel_1 . \tag{6}$$

Then, the bias-corrected $\mathcal{A}$ *-TranSAR* estimator is defined as

$$\widehat{\theta}_{\mathcal{A}\text{-}TranSAR} = \widehat{\omega} + \widehat{\delta}. \tag{7}$$

We summaries the two-stage transfer learning procedure under the SAR model in Algorithm 3.1.

Remark 3. The loss functions ℓ_1 and ℓ_2 in the two stages are allowed to be different. In fact, we make no specific requirements on the form of ℓ_1 as long as the resulting estimator in the transferring stage satisfies the certain convergence rates discussed in Section 3. In particular, we consider a classical 2SLS procedure in the second stage as an example, which has been widely used for the SAR model to deal with the endogeneity. Given an $n_0 \times d$ $(d \geq p+q)$ instrumental variables matrix $Q^{(0)}$ containing $\mathbf{X}^{(0)}$, the first-stage estimator in the 2SLS procedure is

$$\widehat{\pi}_j = \underset{\pi_j \in \mathbb{R}^d}{\arg\min} \left\{ \frac{1}{2n_0} \parallel \mathbb{X}_j^{(0)} - Q^{(0)} \pi_j \parallel_2^2 \right\}, \tag{8}$$

for each $j \in [p+q]$, where $\mathbb{X}^{(0)} = (W_1{}^{(0)} Y^{(0)}, \ldots, W_p{}^{(0)} Y^{(0)}, \mathbf{X}^{(0)}) \in \mathbb{R}^{n_0 \times (p+q)}$, $\mathbb{X}_j^{(0)}$ is the j-th column of $\mathbb{X}^{(0)}$ and π_j's are coefficients of instrumental variables. Denote the fitted values of $\mathbb{X}_j^{(0)}$'s by $\widehat{\mathbb{X}}_j^{(0)} \equiv Q^{(0)} \hat{\pi}_j = P_{Q^{(0)}} \mathbb{X}_j^{(0)}$, and $\widehat{\mathbb{X}}^{(0)} = (\widehat{\mathbb{X}}_1^{(0)}, \ldots, \widehat{\mathbb{X}}_{p+q}^{(0)})$, where the projection matrix is $P_{Q^{(0)}} \equiv Q^{(0)} (Q^{(0)\mathsf{T}} Q^{(0)})^{-1} Q^{(0)\mathsf{T}}$. In the SAR model, instrumental variables could be

$$Q^{(0)} = (\mathbf{X}^{(0)}, W_1^{(0)} \mathbf{X}^{(0)}, \ldots, W_p^{(0)} \mathbf{X}^{(0)})$$

with $d = q(1+p)$ columns. We note that if the first column of $\mathbf{X}^{(0)}$ is the intercept term 1, we need to drop it in $W_j^{(0)} \mathbf{X}^{(0)} (j \in [p])$ to avoid multicollinearity. See Kelejian and Prucha (1998) and Lee (2003) for several alternative choices of instruments. The corresponding loss function for the second stage estimation is

$$\ell_2 \left(\mathfrak{D}^{(0)} \mid \theta \right) = \parallel Y^{(0)} - \widehat{\mathbb{X}}^{(0)} \theta \parallel_2^2 . \tag{9}$$

Then, the bias-corrected $\mathcal{A}$-TranSAR estimator can be equivalently obtained by

$$\widehat{\theta}_{\mathcal{A}\text{-}TranSAR} = \underset{\theta \in \mathbb{R}^{p+q}}{\arg\min} \frac{1}{2n_0} \ell_2 \left(\mathfrak{D}^{(0)} \mid \theta \right) + \lambda_\delta \parallel \theta - \widehat{\omega} \parallel_1 . \tag{10}$$

Remark 4. For high dimensional settings in the reduced form, the number of instruments is larger than the sample size. We could consider using a penalized 2SLS procedure as

$$\widehat{\pi}_j = \underset{\pi_j \in \mathbb{R}^d}{\arg\min} \left\{ \frac{1}{2n_0} \parallel \mathbb{X}_j^{(0)} - Q^{(0)} \pi_j \parallel_2^2 + p_{\lambda_{\pi_j}}(\pi_j) \right\}, \tag{11}$$

for certain penalty function $p_{\lambda_{\pi_j}}(\cdot)$. In the literature, Belloni et al. (2012) proposed the implementation of Lasso and post-Lasso methods to form the first-stage predictions and estimate

optimal instruments in a linear instrumental variables model. Fan and Liao (2014) developed a novel penalized GMM to cope with the endogeneity of a high-dimensional model. Cheng and Liao (2015) studied the problem of choosing valid moments for GMM estimation and developed asymptotic results for high-dimensional GMM shrinkage estimators that allow for nonsmooth sample moments and weakly dependent observations. Zhu (2018) and Gold, Lederer, and Tao (2020) introduced a two-stage Lasso procedure and established some finite-sample convergence rates under different assumptions.

3.2. A Transferable Source Detection Algorithm

The $\mathcal{A}$ -*TranSAR* estimator requires the user-specified set $\mathcal{A}$ to be properly specified. However, in most empirical applications, the "oracle" informative set $\mathcal{A}$ is unknown. Transferring adversarial auxiliary samples may not improve the performance of the target model and could even cause worse results. The effect of adversarial auxiliary samples is also called *negative transfer*, meaning that certain inferior results are caused by uninformative source data (Torrey and Shavlik 2010).

An intuitive approach might involve seeking a metric of similarity between each source and the target data. We compute the initial loss based on the target dataset. Next, we combine the target dataset with each source dataset individually to estimate and compute the corresponding losses. If a source dataset is similar to the target dataset, we anticipate two loss values to be comparably close. Hence, we select a critical value to identify which source datasets are informative. For spatial datasets characterized by a network structure, data splitting used in Tian and Feng (2023) cannot be directly applied since it could disrupt this network integrity. As a result, we introduce the spatial residual bootstrap approach as an alternative. This idea is illustrated in detail as follows.

First, we compute an initial estimator $\widehat{\theta}_{ini} = (\widehat{\lambda}_{ini}^{\intercal}, \widehat{\beta}_{ini}^{\intercal})^{\intercal}$ using the target samples with certain estimation methods, such as 2SLS, QMLE, or GMM. Then, we obtain the residuals $\hat{e} = Y^{(0)} - \mathbb{X}^{(0)}\hat{\theta}_{ini}$, where $\mathbb{X}^{(0)} = (W_1^{(0)}Y^{(0)}, \ldots, W_p^{(0)}Y^{(0)}, \mathbf{X}^{(0)})$. Similar to residual bootstrap, we generate three independent copies, $\widehat{e}^{(1)}$, $\widehat{e}^{(2)}$ and $\widehat{e}^{(3)}$ of sample size n_0 from the empirical distribution of $\widehat{e}$. Then, we generate three-fold bootstrap samples $Y^{(0,r)}$ and $\mathbf{X}^{(0,r)}$ by

$$Y^{(0,r)} = \mathbb{X}^{(0)}\widehat{\theta}_{ini} + \widehat{e}^{(r)}, \quad \mathbf{X}^{(0,r)} = \mathbf{X}^{(0)}, \tag{12}$$

for $r \in [3]$. Let $Z^{(0,r)} = (Y^{(0,r)}, \mathbf{X}^{(0,r)})$. Let $Z^{(0,-r)}$ be the concatenated bootstrap samples without the r-th sample, for example, $Z^{(0,-1)} = (Z^{(0,2)\intercal}, Z^{(0,3)\intercal})^{\intercal}$. Define $\mathbf{X}^{(0,-r)}$, $\mathbb{X}^{(0,r)}$ and $\mathbb{X}^{(0,-r)}$ similarly.

Second, for each $r \in [3]$, we fit the model in (1) on the bootstrap samples $Z^{(0,-r)}$ and use the estimator $\widehat{\theta}^{(0,r)}$ to compute the loss value on the remaining bootstrap sample $Z^{(0,r)}$, that is,

$$\mathcal{L}^{[r]}(\widehat{\theta}^{(0,r)}) \equiv \frac{1}{2n_0} \parallel Y^{(0,r)} - \mathbb{X}^{(0,r)}\widehat{\theta}^{(0,r)} \parallel_2^2 . \tag{13}$$

Then, we compute the average of three loss values, $\mathcal{L}^{(0)} = \sum_{r=1}^{3} \mathcal{L}^{[r]}(\widehat{\theta}^{(0,r)})/3$, which serves as the baseline loss value for the target dataset. We can also compute the standard deviation of three loss values, denoted by

$$\hat{\sigma} = \sqrt{\frac{1}{2}\sum_{r=1}^{3} (\mathcal{L}^{[r]}(\hat{\theta}^{(0,r)}) - \mathcal{L}^{(0)})^2}, \tag{14}$$

which measures the dispersion of bootstrapped loss values $\mathcal{L}^{[r]}(\widehat{\theta}^{(0,r)})$ relative to their average $\mathcal{L}^{(0)}$. Next, for each $r = 1, 2, 3$, we combine $Z^{(0,-r)}$ with each k-th source dataset, fit the model in (1) on the newly combined data and compute the loss denoted by $\mathcal{L}^{(k,r)}$ on $Z^{(0,r)}$. Similarly, the average loss for the k-th source dataset over three-fold bootstrap samples is $\mathcal{L}^{(k)}$. Consequently, the difference between $\mathcal{L}^{(k)}$ and $\mathcal{L}^{(0)}$ provides a metric of the similarity between the k-th source and the target data. Source datasets whose corresponding loss differences are below certain threshold are recognized as transferable informative data. Following the idea of Tian and Feng (2023), we choose $\hat{\sigma} \vee 0.01$ as the detection threshold. Then, the estimated transferable sources $\widehat{\mathcal{A}}$ is

$$\widehat{\mathcal{A}} = \left\{ k = 1, \ldots, K \mid \mathcal{L}^{(k)} - \mathcal{L}^{(0)} \leq (\widehat{\sigma} \vee 0.01) \right\}. \tag{15}$$

Once the transferable set $\mathcal{A}$ is estimated, we can apply the $\mathcal{A}$ -*TranSAR* in Algorithm 3.1 to obtain the transfer learning estimator for the SAR model, denoted by $\widehat{\theta}_{\widehat{\mathcal{A}}\text{-}TranSAR}$, which is termed as the *TranSAR* estimator. The procedure described above is summarized in Algorithm 3.2.

4. Theoretical Results

In this section, we establish an estimation error rate for the $\mathcal{A}$ -*TranSAR* estimator as $\underline{n} \equiv \min_{k\in[K]\cup\{0\}} n_k \to \infty$. Subsection 4.1 is for general results, and Subsection 4.2 is for the 2SLS estimation. Finally, we prove the consistency of the transferable source detection algorithm in Subsection 4.3. For theoretical analysis, we define the population form for the above estimator as

$$\omega^{\mathcal{A}} = \arg\min_{\omega\in\mathbb{R}^{p+q}} \sum_{k\in\mathcal{A}} \alpha_k \bar{\ell}_1^{(k)}(\omega), \tag{16}$$

where $\alpha_k = \frac{n_k}{n_{\mathcal{A}}}$ and $\bar{\ell}_1^{(k)}(\omega)$ is the corresponding population objective function for dataset $\mathfrak{D}^{(k)}$. Under some regularity conditions, the first-stage estimator in Eq. (5) is consistent, that is, $\hat{\omega} - \omega^{\mathcal{A}} \xrightarrow{p} 0$. Analogously, define the "true parameter" based on the single source k as

$$\omega^{(k)} = \arg\min_{\omega\in\mathbb{R}^{p+q}} \bar{\ell}_1^{(k)}(\omega). \tag{17}$$

4.1. Convergence Rate under User-Specified Set $\mathcal{A}$

Assumption 1 (Network structure). The spatial weight matrices $W_l^{(k)}(k \in \{0\} \cup [K], l \in [p])$ are nonstochastic with zero diagonals, and p is fixed and does not depend on n_0 or n_k. $S^{(k)}(\lambda) = I_{n_k} - \sum_{l=1}^{p} \lambda_l^{(k)} W_l^{(k)}$ is invertible for all $\lambda \in \Lambda$, where the parameter space Λ is compact. Matrices $W_l^{(k)}$ and $S^{(k)}(\lambda)^{-1}$ are uniformly bounded in both row and column sum norms for all $n_k, k \in [K] \cup \{0\}, l \in [p]$ and $\lambda \in \Lambda$. And for all large enough

Algorithm 2 TranSAR

Input: Target: $\mathfrak{D}^{(0)}$; sources: $(\mathfrak{D}^{(k)}, k \in [K])$; tuning parameter: $\lambda_\theta^{(k,r)}, k \in [K] \cup \{0\}, r \in [3]$; initial estimator, $\widehat{\theta}_{ini}$.

Output: $\widehat{\theta}_{TranSAR}$ and $\widehat{\mathcal{A}}$.

Residuals: $\hat{e} = Y^{(0)} - \mathbb{X}^{(0)}\hat{\theta}_{ini}$;

for r *from* 1 *to* 3 **do**

  Generate copy $\hat{e}^{(r)}$ from $\hat{e}$ and $Y^{(0,r)}$, $\mathbf{X}^{(0,r)}$, $Z^{(0,-r)}$, $\mathbf{X}^{(0,-r)}$, $\mathbb{X}^{(0,r)}$ and $\mathbb{X}^{(0,-r)}$;

  $\widehat{\theta}^{(0,r)} \leftarrow \underset{\theta \in \mathbb{R}^{p+q}}{\arg\min} \frac{1}{2n_0} \ell_2\left(Z^{(0,-r)} \mid \theta\right) + \lambda_\theta^{(0,r)} \|\theta\|_1$;

  for k *from* 1 *to* K **do**

    $\widehat{\theta}^{(k,r)} \leftarrow \underset{\theta \in \mathbb{R}^{p+q}}{\arg\min} \left\{ \frac{1}{2(n_k+2n_0)} \left\{\ell_2(\mathfrak{D}^{(k)} \cup Z^{(0,-r)} \mid \theta)\right\} + \lambda_\theta^{(k,r)} \|\theta\|_1 \right\}$;

  end

end

Calculate the average losses: $\mathcal{L}^{(k)} = \frac{1}{3}\sum_{r=1}^{3} \mathcal{L}^{[r]}(\hat{\theta}^{(k,r)}), k = 0, 1, \ldots, K$;

Calculate the standard deviation: $\hat{\sigma} \leftarrow \sqrt{\frac{1}{2}\sum_{r=1}^{3} \left(\mathcal{L}^{[r]}(\hat{\theta}^{(0,r)}) - \mathcal{L}^{(0)}\right)^2}$;

The estimated transferable sources: $\hat{\mathcal{A}} \leftarrow \left\{k = 1, \ldots, K \mid \mathcal{L}^{(k)} - \mathcal{L}^{(0)} \le (\hat{\sigma} \vee 0.01)\right\}$;

$\widehat{\theta}_{TranSAR}$ is obtained by Algorithm 1 via letting $\mathcal{A} = \hat{\mathcal{A}}$;

$n_k, \| W_l^{(k)} S^{(k)}(\lambda)^{-1} \|_F / \sqrt{n_k}$ is bounded away from zero and infinity uniformly over $k \in [K] \cup \{0\}, l \in [p]$ and $\lambda \in \Lambda$.

Assumption 2 (Regular conditions for the design matrix and error terms).

(1) Denote $\mathbf{X}^{(0)} \equiv (\mathbf{x}_1^{(0)}, \ldots \mathbf{x}_{n_0}^{(0)})^\intercal$ and the covariance matrix of $\mathbf{x}_i^{(0)}$ for each i is Σ_0. For any $\beta \in \mathbb{R}^q, \mathbf{x}_i^{(0)\intercal}\beta$ $(i \in [n_0])$, are *i.i.d.* $\kappa_0 \| \beta \|_2$ sub-Gaussian with mean zero, where κ_0 is a positive constant, and $0 < \underline{\psi_0} = \lambda_{\min}(\Sigma_0) \le \lambda_{\max}(\Sigma_0) = \overline{\psi_0} < \infty$.

(2) The error terms $V_i^{(0)}$ $(i \in [n_0])$ are *i.i.d.* sub-Gaussian with mean zero and variance σ^2.

(3) As $n_0 \to +\infty$, $\mathbb{P}\left(\lambda_{\min}(\frac{(\mathbb{X}^{(0)})^\intercal \widehat{\mathbb{X}}^{(0)}}{n_0}) \ge \phi_0\right) \to 1$, where ϕ_0 is a positive constant.

Assumption 3 (Convergence rates of the first step estimator). The first step estimator $\widehat{\omega}$ minimizing (5) satisfies $\| \widehat{\omega} - \omega^{\mathcal{A}} \|_1 = O_{\mathbb{P}}(a_{n_{\mathcal{A}}}^{(1)})$, and $n_{\mathcal{A}}^{-1} \| \mathbb{X}^{\mathcal{A}}(\widehat{\omega} - \omega^{\mathcal{A}}) \|_2^2 = O_{\mathbb{P}}(a_{n_{\mathcal{A}}}^{(2)})$, where $a_{n_{\mathcal{A}}}^{(1)} \to 0$ and $a_{n_{\mathcal{A}}}^{(2)} \to 0$ as $n_{\mathcal{A}} \to \infty$, and $\mathbb{X}^{\mathcal{A}}$ is stack of $\mathbb{X}^{(k)}$ for $k \in \mathcal{A}$.

Assumption 4 (Instruments).

When n_0 is large enough, the IV matrix

$$Q^{(0)} = \left(\mathbf{X}^{(0)}, W_1^{(0)}\mathbf{X}^{(0)}, \ldots, W_q^{(0)}\mathbf{X}^{(0)}\right) \tag{18}$$

satisfies $0 < \psi_*^{(0)} \le \frac{1}{\sqrt{n_0}} \| Q^{(0)} \|_2 \le \psi^{(0)*} < \infty$ for some constants $\psi_*^{(0)}$ and $\psi^{(0)*}$. There exist constant coefficient vectors $\pi_j \in \mathbb{R}^d$ for $j \in [p+q]$ such that $\max_{j=1,\ldots,p+q} \| \widehat{\pi}_j - \pi_j \|_2 = O_{\mathbb{P}}(\sqrt{\log(d)/n_0})$ and $\| \pi_j \|_2^2$ is uniformly bounded over j and n_0, where $\widehat{\pi}_j$ $(j \in [p+q])$ are defined in Eq. (8).

Remark 5. Assumption 1 is standard in spatial econometrics, see Kelejian and Prucha (1999) and Lee (2004). Meanwhile, we assume the number of spatial weight matrices p is fixed, as either one or very few spatial weight matrices are often used in most empirical studies. Assumption 2 requires the predictors and error terms to be sub-Gaussian, which is standard in the literature of transfer learning, such as Li, Cai, and Li (2022b) and Tian and Feng (2023). The eigenvalue condition in Assumption 2 (iii) is widely used in high-dimensional statistical literature or can be derived from various regular conditions (Loh and Wainwright 2012; Raskutti, Wainwright, and Yu 2010; van de Geer and Bühlmann 2009). Assumption 3 imposes conditions on the convergence rate of the first step estimator, which will be verified using a concrete example in Subsection 4.2. Assumption 4 is about the validity of the instrument matrix $Q^{(0)}$ and the refitted value $\widehat{\mathbb{X}}^{(0)}$.

Theorem 1. **(Convergence rate of the $\mathcal{A}$-TranSAR under user-specified set $\mathcal{A}$).** Let the parameter space be $\Theta = \Lambda \times \{\beta \mid\mid \beta \|_2 < M\}$ for some positive constant M. Under *Assumptions 1* - 4, if the loss function ℓ_1 is twice differentiable, $\sup_{k \in \mathcal{A}} \| \Sigma^{\mathcal{A}^{-1}} \Sigma^{(k)} \|_1 = \check{C} < \infty$, $s\frac{\log q}{n_0} = o(1)$, $q \asymp d$, and $\lambda_\delta \gtrsim \sqrt{\log(q)/n_0}$, where

$$\Sigma^{(k)} \equiv \int_0^1 \frac{\partial^2}{\partial\theta\partial\theta^\intercal} \bar{\ell}_1^{(k)}(\theta_0 + t(\omega^{(k)} - \theta_0))dt, \; k \in \mathcal{A}, \tag{19}$$

$$\Sigma^{\mathcal{A}} \equiv \sum_{k \in \mathcal{A}} \int_0^1 \alpha_k \frac{\partial^2}{\partial\theta\partial\theta^\intercal} \bar{\ell}_1^{(k)}(\theta_0 + t(\omega^{\mathcal{A}} - \theta_0))dt, \alpha_k = \frac{n_k}{n_{\mathcal{A}}}. \tag{20}$$

Denote $S = \text{support}(\theta_0)$ and $s = |S|$. Then $\widehat{\theta}_{\mathcal{A}\text{-}TranSAR}$ defined in Eq. (7) satisfies

$$\| \widehat{\theta}_{\mathcal{A}\text{-}TranSAR} - \theta_0 \|_2^2 = O_{\mathbb{P}}\left((\sqrt{\frac{\log q}{n_0}} h_{\mathcal{A}}) \wedge (s\frac{\log q}{n_0}) \wedge h_{\mathcal{A}}^2 + a_{n_{\mathcal{A}}}^{(2)}\right) \tag{21}$$

where $h_{\mathcal{A}}$ is the tolerance level for the user-specified set $\mathcal{A}$ defined in Eq. (4).

Remark 6. When we consider the case of fixed dimension of covariates q, under some regularity conditions, $a^{(2)}_{n_{\mathcal{A}}} = \frac{1}{n_{\mathcal{A}}}$. Then, we could obtain $\| \widehat{\theta}_{\mathcal{A}\text{-}TranSAR} - \theta_0 \|_2^2 = O_{\mathbb{P}}(\min(n_0^{-1}, h_{\mathcal{A}}^2) + n_{\mathcal{A}}^{-1})$, which implies that when the sample size $n_{\mathcal{A}}$ of the informative source data is large enough and $h_{\mathcal{A}}$ is small enough, the estimation convergence rate of the $\mathcal{A}$-TranSAR estimator is better than that only using the target data. And if $h_{\mathcal{A}}$ is too large, the convergence rate of the $\mathcal{A}$-TranSAR estimator may be worse than that only using the target data, that is, negative transfer.

4.2. Convergence Rate for the 2SLS Loss under User-Specified Set $\mathcal{A}$

In this section, we verify the requirements in the above theorem for the 2SLS estimation and simplify the convergence rate of the $\mathcal{A}$ *-TranSAR* estimator. As in Eq. (9), let $\ell_1(\cdot \mid \omega)$ be the loss function of the 2SLS. Let $Y^{\mathcal{A}}$ denote the stack consisting of $Y^{(k)}$ for all $k \in \mathcal{A}$, that is, $Y^{\mathcal{A}} = (Y^{(k)\mathsf{T}}, k \in \mathcal{A})^{\mathsf{T}}$. $\mathbf{X}^{\mathcal{A}}$ and $\mathbb{X}^{\mathcal{A}}$ are defined similarly. Then the objective function (5) could be rewritten as

$$\frac{1}{2n_{\mathcal{A}}} \| Y^{\mathcal{A}} - \widehat{\mathbb{X}}^{\mathcal{A}}\omega \|_2^2 + \lambda_\omega \| \omega \|_1, \tag{22}$$

where $\widehat{\mathbb{X}}^{(k)}$ is defined similarly as $\widehat{\mathbb{X}}^{(0)}$ in Subsection 3.1, and $\widehat{\mathbb{X}}^{\mathcal{A}}$ is their stack. Here the IV matrices are $Q^{(k)}$ and $Q^{\mathcal{A}}$. We impose several similar conditions on $\mathfrak{D}^{(k)}$ ($k \in \mathcal{A}$) in the following.

Assumption 5. ***(Regular conditions for design matrices and error terms on the k-th source).*** For source $k \in \mathcal{A}$, we assume: (i) For the design matrix $\mathbf{X}^{(k)} = (\mathbf{x}_1^{(k)}, \ldots \mathbf{x}_{n_k}^{(k)})^{\mathsf{T}}$ and any $\beta \in \mathbb{R}^q$, $\mathbf{x}_i^{(k)\mathsf{T}}\beta$ ($i \in [n_k]$) are *i.i.d.*$\kappa_k \| \beta \|_2$ sub-Gaussian with mean zero for constant κ_k, where $\max_{k\in\mathcal{A}} \kappa_k < \infty$. (ii) The error terms $V_i^{(k)}$ ($i \in [n_k]$) are *i.i.d.* sub-Gaussian with mean zero and variance σ_k^2. (iii) As $n_k \to +\infty$, $\mathbb{P}\left(\lambda_{\min}\left(\frac{\widehat{\mathbb{X}}^{(k)\mathsf{T}}\widehat{\mathbb{X}}^{(k)}}{n_k}\right) \geq \phi_k\right) \to 1$, where ϕ_k is a positive constant.

Assumption 6 (Instruments for the k-th source). For the instruments

$$Q^{(k)} = \left(\mathbf{X}^{(k)}, W_1^{(k)}\mathbf{X}^{(k)}, \ldots, W_q^{(k)}\mathbf{X}^{(k)}\right), \ k \in \mathcal{A} \cup \{0\}, \tag{23}$$

when n_k, the number of rows of $Q^{(k)}$, is large enough, $0 < \psi_*^{(k)} \leq \frac{1}{\sqrt{n_k}} \| Q^{(k)} \|_2 \leq \psi^{(k)*} < \infty$ for some constants $\psi_*^{(k)}$ and $\psi^{(k)*}$. There exist constant coefficient vectors $\pi_j^{(k)} \in \mathbb{R}^d$ for $j \in [p+q]$ such that $\max_{j=1,\ldots,p+q} \| \hat{\pi}_j - \pi_j^{(k)} \|_2 = O_{\mathbb{P}}(\sqrt{\log(d)/n_k})$ and $\| \pi_j^{(k)} \|_2$ is uniformly bounded over j and k.

Under Assumptions 5 and 6, the convergence rate of the $\mathcal{A}$ -TranSAR estimator is summarized in the following Theorem 2.

Theorem 2. If *Assumptions 1, 2, 4, 5, and 6* hold, $\sup_{n_k} \mathbf{B}^{(k)} < \infty$ with $\mathbf{B}^{(k)} \equiv 1 \vee \| \beta_0^{(k)} \|_2$, K is a fixed, $q \asymp d$, $s\frac{\log q}{n_0} = o(1)$, $s^2\frac{\log n_0}{n_0} = o(1)$, $n_{\mathcal{A}} > n_0$, $q^2 h = O(1)$, $\lambda_\delta \gtrsim \sqrt{(\log q)/n_0}$ and $\lambda_\omega \gtrsim \sqrt{(\log q)/n_{\mathcal{A}}}$, where $S = \text{support}(\theta_0)$ and $s = |S|$, then we have

$$\| \widehat{\theta}_{\mathcal{A}\text{-}TranSAR} - \theta_0 \|_2^2 = O_{\mathbb{P}}\left(\left(\sqrt{\frac{\log q}{n_0}}h_{\mathcal{A}}\right) \wedge \left(\frac{s\log q}{n_0}\right) \wedge h_{\mathcal{A}}^2 + \left(\frac{\log n_A}{n_{\mathcal{A}}}h_{\mathcal{A}}^2\right) \wedge \frac{\log^3 n_A}{n_{\mathcal{A}}^3} + \frac{s\log q}{n_{\mathcal{A}}}\right). \tag{24}$$

Theorem 2 indicates that if $h_{\mathcal{A}} \ll s\sqrt{\log q/n_0}$ and $n_{\mathcal{A}} \gg n_0$, the $\mathcal{A}$ *-TranSAR* estimator is better than the classical 2SLS estimator based on only the target data, which demonstrates the usefulness of transfer learning in the SAR models.

4.3. Detection Consistency

Finally, we will show that our proposed transferable source detection algorithm is consistent even for a high dimensional model, that is, $q \gg n_0$. Here we consider the pre-specified informative set $\mathcal{A}$ satisfies $h_{\mathcal{A}} = o(1)$, and other identifiability conditions:

Assumption 7 (Identifiability condition). The tolerance level $h_{\mathcal{A}} = o(1)$ as the minimal sample size $\underline{n} \equiv \min_{k\in[K]\cup\{0\}} n_k \to \infty$. For any $k \in [K]\cup\{0\}$ and $r \in [3]$, $q\log(q) \| \tilde{\theta}^{(k)} - \widehat{\tilde{\theta}}^{(k,r)} \|_1 = o_{\mathbb{P}}(1)$, where $\tilde{\theta}^{(k)}$ is defined in Eq. (C.34). For $k \in \mathcal{A}^c \equiv [K] - \mathcal{A}$, we control the signal power as

$$\underline{\lambda} \| \theta_0 - \tilde{\theta}^{(k)} \|_2^2 \geq \| \theta_0 - \tilde{\theta}^{(k)} \|_1 + \bar{C}\left[(\| \theta_0 - \widehat{\theta}^{(0,r)} \|_1) \vee 1\right] - \frac{\sigma^2}{2n_0}\text{tr}\{[(S(\tilde{\lambda}^{(k)}) - S)S^{-1}]^{\mathsf{T}}[(S(\tilde{\lambda}^{(k)}) + S)S^{-1}]\}, \tag{25}$$

and the gap between θ_0 and $\tilde{\theta}^{(k)}$ for $k \in \mathcal{A}^c$,

$$\varphi_k \equiv \sqrt{\frac{1}{n_0}}(\| \theta_0 - \tilde{\theta}^{(k)} \|_1 + 1)^2 = o(1), \tag{26}$$

where $\bar{C} > 0$ is a constant and $\underline{\lambda} \equiv \liminf_{n_0\to\infty} \lambda_{\min}(\frac{1}{2n_0}\mathbb{E}\mathbf{X}^{\star\mathsf{T}}\mathbf{X}^{\star}) > 0$, with $\mathbf{X}^{\star} \equiv (\mathbf{W}^{(0)}(\mathbf{I}_p \otimes (S^{-1}\mathbf{X}^{(0)}\beta_0)), \mathbf{X}^{(0)})$.

Remark 7. Conditions (7) and (26) guarantee a suitable gap between the population-level coefficients in set $\mathcal{A}^c$ and the true coefficient of the target model. Intuitively, assuming an appropriate signal strength is necessary to ensure the consistency of estimators. On the one hand, a very weak signal makes it challenging for the algorithm to differentiate informative and uninformative sources. On the other hand, an excessively strong signal can lead the first step estimator to mimic the model structure of the source, thereby hindering accurate identification. And we also require the consistency of these estimators, which is guaranteed under some regularity conditions. Since Lasso and 2SLS estimators are still consistent under heteroscedasticity (Jia, Rohe, and Yu 2013; Kelejian and Prucha 2010; Lin and Lee 2010; Sato and Matsuda 2017; Wagener and Dette 2013), our detection consistency is also guaranteed under heterogeneous noise levels across source datasets.

Next, we establish the detection consistency of $\widehat{\mathcal{A}}$ in the following theorem.

Theorem 3 (Detection consistency of $\widehat{\mathcal{A}}$). If *Assumptions (1)-(4)* and *(7)* are satisfied, $\| \beta_0 \|_2 < \infty$ and $\frac{\log q}{n_0} = o(1)$, then the automatically detected transferable index set $\widehat{\mathcal{A}}$ in *Eq. (15)* is consistent, that is,

$$\mathbb{P}(\widehat{\mathcal{A}} = \mathcal{A}) \to 1, \text{as } \underline{n} \to \infty. \tag{27}$$

Theorem 3 shows that the proposed transferable source detection algorithm is consistent under a weak conditional $h_{\mathcal{A}} = o(1)$. And further, if the transferable sources are unknown, we could have the same convergence rate of the $\widehat{\theta}_{TranSAR}$ as $\widehat{\theta}_{\mathcal{A}\text{-}TranSAR}$ in Theorem 1 for user-specified set $\mathcal{A}$ as follows.

Corollary 1. Under the same conditions as in Theorem 3, the estimator $\widehat{\theta}_{TranSAR}$ maintains the same convergence rate as $\widehat{\theta}_{\mathcal{A}\text{-}TranSAR}$ for user-specified set $\mathcal{A}$ in Theorem 1, that is,

$$\| \widehat{\theta}_{TranSAR} - \theta_0 \|_2^2 = O_{\mathbb{P}} \left(\left(\sqrt{\frac{\log q}{n_0}} h_{\mathcal{A}} \right) \wedge \left(s \frac{\log q}{n_0} \right) \wedge h_{\mathcal{A}}^2 + a_{n_{\mathcal{A}}}^{(2)} \right). \tag{28}$$

However, the assumption $h_{\mathcal{A}} = o(1)$ alone does not guarantee protection against negative transfer. We will next demonstrate that our proposed transferable source detection algorithm achieves "safe" transfer under appropriate regularity conditions.

4.4. Safe Transfer Learning

In this section, we will show that the *TranSAR* estimator is "safe" under a weak condition.

Theorem 4 (Safe transfer learning). *Under Assumptions (1)-(4) and* $h = o(1)$, $a_{n_{\mathcal{A}}}^{(1)} \vee a_{n_{\mathcal{A}}}^{(2)} = O_{\mathbb{P}}(s\sqrt{\log q / n_0})$, *if* $\mathbb{P}(\widehat{\mathcal{A}} = \mathcal{A}) \to 1$ *then we have the converge rate of the TranSAR estimator is not slower than the estimator based on only the target data, that is*

$$\hat{R}(\widehat{\theta}_{TranSAR}) = O_{\mathbb{P}}(\hat{R}(\widehat{\theta}_{SAR})), \tag{29}$$

where $\hat{R}(\widehat{\theta}) = \| \widehat{\theta} - \theta_0 \|_2^2$.

Remark 8. The intuitions behind Theorem 4 is that if $\mathbb{P}(\widehat{\mathcal{A}} = \mathcal{A}) \to 1$ by Algorithm 3.2, which is verified in Theorem 3, then the TranSAR estimator will perform not worse than the estimator based on only the target data. In Algorithm 3.2, we use $\hat{\sigma}$ to control the loss difference between the [k]-TranSAR estimator and the [0]-TranSAR estimator. This control ensures the convergence rate of h, which is the key in safe transfer learning. And we could also refine the result by using a sharper threshold. More details can be found in the proof of Theorem 4 in supplementary material. It provides a theoretical guarantee for the TranSAR estimator to be "safe" only requiring the detection consistency of the transferable source.

5. Simulation

In this section, we design several simulation experiments to investigate the performance of the proposed method (denoted as *TranSAR*). We compare the finite sample performance of our estimator with the classical 2SLS estimation (which is fitted only using the target dataset, denoted as *SAR*), the $\mathcal{A}$ *-TranSAR* estimator (which is fitted based on the true informative index set $\mathcal{A}$, denoted as *Oracle TranSAR*), pooled estimator (which is fitted using all the source datasets as the transferable sets, denoted as $[K]$ *-TranSAR*), and the generalized linear model transfer learning algorithm (denoted as *glmtrans*) proposed by Tian and Feng (2023) in Experiment 1. The *glmtrans* ignores the spatial information and can be considered as a baseline estimator to show that it is necessary to consider the spatial correlations if they exist. And we examine the sensitivity of the *TranSAR* estimator to heterogeneous noise in Experiment 2. Further, we consider the sensitivity of the *TranSAR* estimator to the spatial effect and the number of spatial matrices in Experiment 3 and the of robustness to the potential misspecification of the spatial weight matrices in Experiment 4, both placed in supplementary material. All experiments are conducted with 100 replications.

Experiment 1: Comparison with existing methods. We generate the data according to Models (1) and (2). In these models, $n_0 = 256, n_1 = n_2 = \cdots = n_K = 100, p = 1, q = 200, K = 20$. The covariates $\mathbf{x}_i^{(k)}$ are generated by multivariate normal distribution with covariance matrix $\Sigma \in \mathbb{R}^{q \times q}$, whose (j, j')-th entry is denoted as $\Sigma_{j,j'}$ for all $0 \le k \le K$. We consider three designs for Σ: (1) $\Sigma_{j,j'} = 1$ for $j = j'$ and $\Sigma_{j,j'} = 0$ otherwise, (2) $\Sigma_{j,j'} = 0.5^{|j-j'|}$, (3) $\Sigma_{j,j'} = 0.9^{|j-j'|}$. We consider two different distributions for the i.i.d. error terms: (1) the standard normal distribution $N(0, 1)$, and (2) the t_2 distribution. The spatial units are assumed to be located on a square grid, and we set up several candidate spatial weight matrices as follows: (1) the first W_1 is the spatial weight matrix where spatial units interact with their neighbors on the left and right sides; (2) the second W_2 is the spatial weight matrix where spatial units interact with their neighbor above and below; (3) the rest of the spatial weight matrices are assumed to be matrices where spatial units interact with their second-nearest, third-nearest, and so on, denoted as $W_3, \ldots, W_N$, where $N < \sqrt{n_0}$. In each dataset, we randomly draw spatial weight matrices from these candidate matrices. The weight matrices $\{W_k^{(l)}\}_{k \in \{0\} \cup [K], l \in [p]}$ are drawn randomly from $\{W_1, \ldots, W_N\}$ without replacement. In Models (1) and (2), we consider coefficients settings as follows. In the target Model (1), for the spatial spillover effects parameter λ_0, we set: $\lambda_0 = 0.4$ if $p = 1$, $\lambda_0 = (0.4, 0.3)^\intercal$ if $p = 2$, $\lambda_0 = (0.4, 0.3, \mathbf{0}_{p-2}^\intercal)^\intercal$ if $p > 2$, and the regression coefficients $\beta_0 = (\mathbf{1}_3^\intercal, \mathbf{0}_{q-3}^\intercal)^\intercal \in \mathbb{R}^q$. In Model (2),

$$\lambda_0^{(k)} = \begin{cases} \lambda_0, & \text{if } k \in \mathcal{A}, \\ -\lambda_0, & \text{if } k \in \mathcal{A}^c, \end{cases} \tag{30}$$

$$\beta_{j0}^{(k)} = \beta_{j0} - 0.051\{j \in \mathcal{H}_k, k \in \mathcal{A}\} - 21\{j \in \mathcal{H}_k, k \in \mathcal{A}^c\}, \quad j = 1, \ldots, q, \tag{31}$$

where $\mathcal{H}_k$ is a random subset of $[q]$ with $|\mathcal{H}_k| = H$ if $k \in \mathcal{A}$ (i.e., H is the number of different coefficients in the informative set), or $|\mathcal{H}_k| = \frac{q}{2}$ if $k \notin \mathcal{A}$. So, on the informative datasets, the

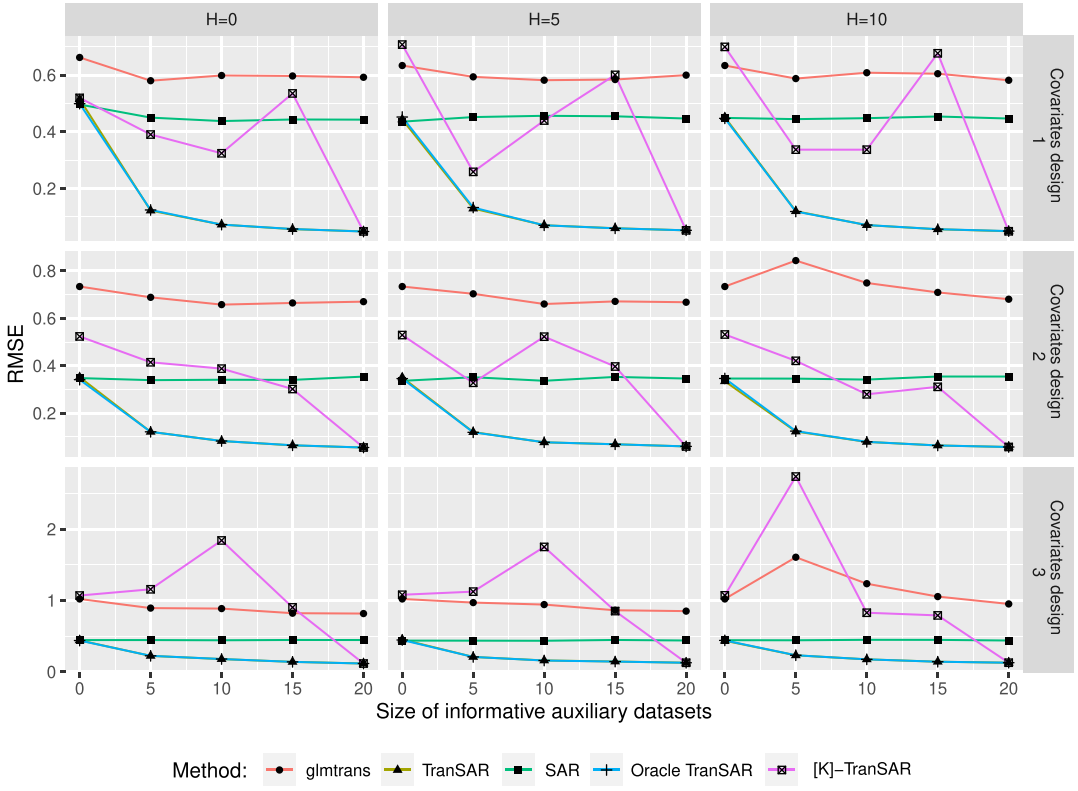


Figure 2. RMSE of three types of covariates designs with normal error distribution with $H = 0, 5, 10$.

differences of different $\beta_{j0}^{(k)}$'s are 0.05; on uninformative datasets, the ones of different $\beta_{j0}^{(k)}$'s are 2.

We compute the root mean squared errors (RMSE), RMSE $= \sqrt{\frac{1}{R}\sum_{r=1}^{R} \| \theta_0 - \widehat{\theta}^{[r]} \|^2}$, where $\widehat{\theta}^{[r]}$ is the estimator of the r-th replication, based on the 2SLS loss with $q = 200$ and display them under different settings in Figure 2. In order to demonstrate the advantages of transfer learning, we let the size of informative auxiliary datasets $|\mathcal{A}|$ increase from 0 to 20 to show the decreasing trend in RMSE. As shown in Figure 2, the *TranSAR* and *Oracle TranSAR* can significantly reduce the RMSE, resulting in more precise estimation. Additionally, we can see that the *TranSAR* performs almost the same as the oracle transfer learning method *Oracle TranSAR*. That means, our proposed detection method is able to select informative sets with a high probability. Moreover, from Figure 2, when the number of informative sets is small, the *negative transfer* effect of the pooled method, [K] -*TranSAR* becomes apparent. Hence, it is necessary to select informative sets accurately. The same conclusion applies to all three types of design matrices and the three different settings for H. We show the detailed and comprehensive simulation results in supplementary material.

Experiment 2: Sensitivity to heterogeneous noise. We consider the same data generating process as in Experiment 1 with covariates design type 1 and informative set size $|\mathcal{A}| = 10$, but heterogeneous error terms. We set the variance of the error term in the target dataset to be 1 and set an experiment parameter `dispersion` as the measure of variance heterogeneity across datasets. The `dispersion` parameter controls the total magnitude of variance differences between the target and source datasets. Formally, let $\sigma^2 = 1$ be the variance of the target dataset, and σ_k^2 be the variance of the k-th source dataset. The parameter `dispersion`, denoted as Δ, is defined by $\sum_{k=1}^{K} |\sigma_k^2 - \sigma^2| = K \cdot \Delta$, where K is the number of source datasets. Specifically, for each source dataset, a random variance factor is assigned such that the sum of absolute differences between these factors and the target variance σ^2 (which is 1) equals $K \cdot$ `dispersion`, where K is the number of source datasets. For each source, the direction of variance difference (whether larger or smaller than the target) is randomly determined. To examine the relationship between our proposed method and noise heterogeneity, we vary the `dispersion` parameter across different values from 0 to 20. When `dispersion` equals 0, all datasets have identical error variance; as `dispersion` increases, the heterogeneity in error variance across datasets becomes more significant. We present the results in Figure 3. From the results, we can conclude that the Rand Index (RI), Precision, Recall, False Positive Rate (FPR) and False Negative Rate (FNR)[1] of the proposed method are close

[1]The Rand Index is a measure of the similarity between two clustering (here, meaning informative and noninformative set): Let $\mathcal{A}$ be the true set of informative sources and $\hat{\mathcal{A}}$ be the estimated set. The Rand Index is defined as $RI = \frac{a+b}{\binom{K}{2}}$, where K is the number of source datasets, a is the number of pairs of sources that are in the same subset in both partitions (i.e., both in $\mathcal{A}$ or both in $\mathcal{A}^c$, and similarly for $\hat{\mathcal{A}}$), and b is the number of pairs of sources that are in different subsets in both partitions. Precision is defined as $|\mathcal{A} \cap \hat{\mathcal{A}}|/|\hat{\mathcal{A}}|$. Recall is defined as $|\mathcal{A} \cap \hat{\mathcal{A}}|/|\mathcal{A}|$. False Positive Rate is defined as $|\hat{\mathcal{A}} \setminus \mathcal{A}|/|\mathcal{A}^c|$. False Negative Rate is defined as $|\mathcal{A} \setminus \hat{\mathcal{A}}|/|\mathcal{A}|$. We report the averaged metrics over 100 replications.

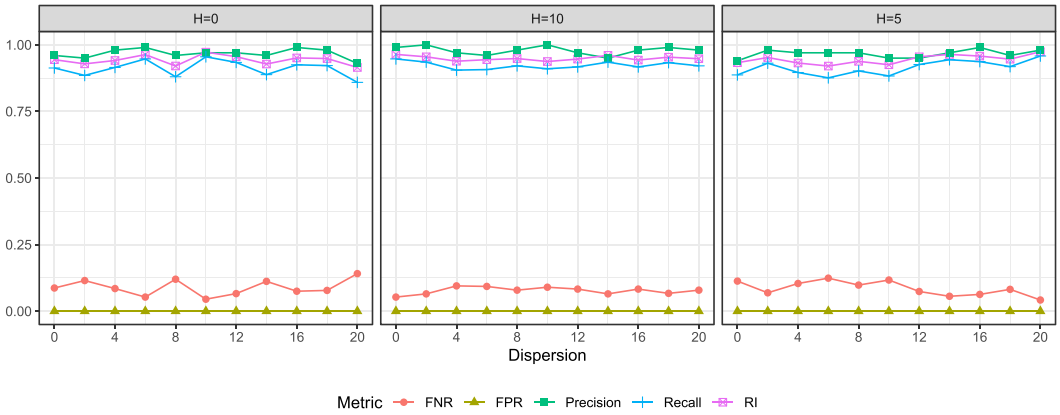


Figure 3. The Rand index, Precision, Recall, FNR and FPR of the proposed method with increasing `dispersion`, and $H = 0, 5, 10$.

Table 1. Average prediction RMSE of the predicted County-level rates of support in the 2020 presidential election.

	Florida	Georgia	Michigan	Minnesota	North Carolina	Ohio
SAR	0.10723	0.10995	0.10494	0.19211	0.10118	0.10984
tranSAR	0.09417	0.09064	0.08033	0.09646	0.10966	0.08297

to 1, 1, 1, 0 and 0 respectively with increasing `dispersion`, indicating that the proposed method is not sensitive to noise heterogeneity.

6. An Application to U.S. Presidential Election Prediction

In this section, we apply the proposed method in Algorithm 3.2 to empirical data from the U.S. presidential election. As shown in Section 2.1, the response variable $Y^{(k)}$ is the difference of support rates in the county level, and covariates $\mathbf{X}^{(k)}$ include voting data of last election, demographic and geographic data, such as including total population, total housing units, poverty people population, etc. With the observed spatial weight matrix $\mathbf{W}^{(k)}$, each swing state is regarded as a target with $k = 0$, and all the other states are considered as sources $k \in [47]$. The target model (1) and sources models (2) are formulated as We run the proposed estimation method, *TranSAR*, using the polls from the 2016 U.S. presidential election and some other demographic data with geographic information. We also estimate the SAR model with Lasso penalty (due to high dimensional covariates) using only the target datasets (denoted as SAR) to compare the empirical performance. Due to the inherent stochastic nature of the resampling procedure in the algorithm, we conduct multiple replications (20 times) for both methods to obtain a stable result. The average prediction RMSE of the county-level vote results of the county-level 2020 U.S. presidential election is compared at Table 1, where RMSE for a *state* is defined as $\sqrt{\sum_{i\in\text{state}}(\hat{Y}_i^{2020} - Y_i^{2020})^2/n_{\text{state}}}$, Y_i^{2020} is the true difference of county-level support rates in the 2020 election, $\hat{Y}_i^{2020}$ is the predicted difference, and n_{state} is the number of counties in the state. For all swing states, the proposed *TranSAR* method performs better than the traditional SAR in terms of RMSE, which demonstrates the necessity of transfer learning in the SAR framework.

Moreover, we also consider the practical rules of the U.S. presidential election, which operate on a "winner-takes-all" basis. In this context, the winner party secures all the state's electoral votes. Given the "winner-take-all" framework of the U.S. presidential election, we focus on predicting the winner party in each state. Leveraging the county-wise population, we compute the final difference of support in the state-level endorsement for presidential contenders belonging to the respective parties. We calculate the rate of support for state-level elections using the following formula: Rate of state-level support $= \sum_i \frac{\text{Rate of county } i \text{ level support} \times \text{Votes of the county } i}{\text{Total votes}}$. Then, we compare the prediction rate of state-level support and the true one in 2020 election, showing in Table 2. It demonstrates that the transfer learning under the SAR framework is able to effectively improve the state-level prediction.

The calculated ratios of correct predictions out of the 20 election result forecasts are collected in Table 3. As shown in Table 3, the method we proposed exhibits better performance, especially in predicting election outcomes in Michigan and Minnesota. Simultaneously, it is noteworthy that for the election predictions in Georgia, neither of the methods works well. It can be explained by the historical context: from 1972 to 2016, Georgia consistently supported the Republican Party, except for Democratic candidates from the southern region. However, the state has become an increasingly competitive battleground. In 2020, the Democratic candidate, Joe Biden, secured a victory over Donald Trump by a narrow margin of 0.2%. This marks a significant departure from history, as the Democrats secured the presidential election with a slim advantage. Evidently, Georgia has become an outlier in this regard. It is evident that predicting the ultimate election outcome for a single state is significantly more challenging than forecasting the rate of state-level support. Nevertheless, we have made commendable progress in this regard.

Next, we attempt to visualize transferable states, conducting a statistical analysis of the results after multiple repetitions. For a

Table 2. Prediction bias of the predicted state-level 2020 election results.

	Florida	Georgia	Michigan	Minnesota	North Carolina	Ohio
SAR	−0.01989	−0.08110	−0.07157	−0.04189	−0.01782	−0.05765
tranSAR	0.00546	−0.04186	0.03522	−0.02228	−0.02163	−0.03189

Table 3. The ratios of correct predictions for election result of 2020 election using the 2016 election data as training data.

	Florida	Georgia	Michigan	Minnesota	North Carolina	Ohio
SAR	100%	0%	35%	90%	100%	100%
tranSAR	100%	0%	100%	100%	100%	100%

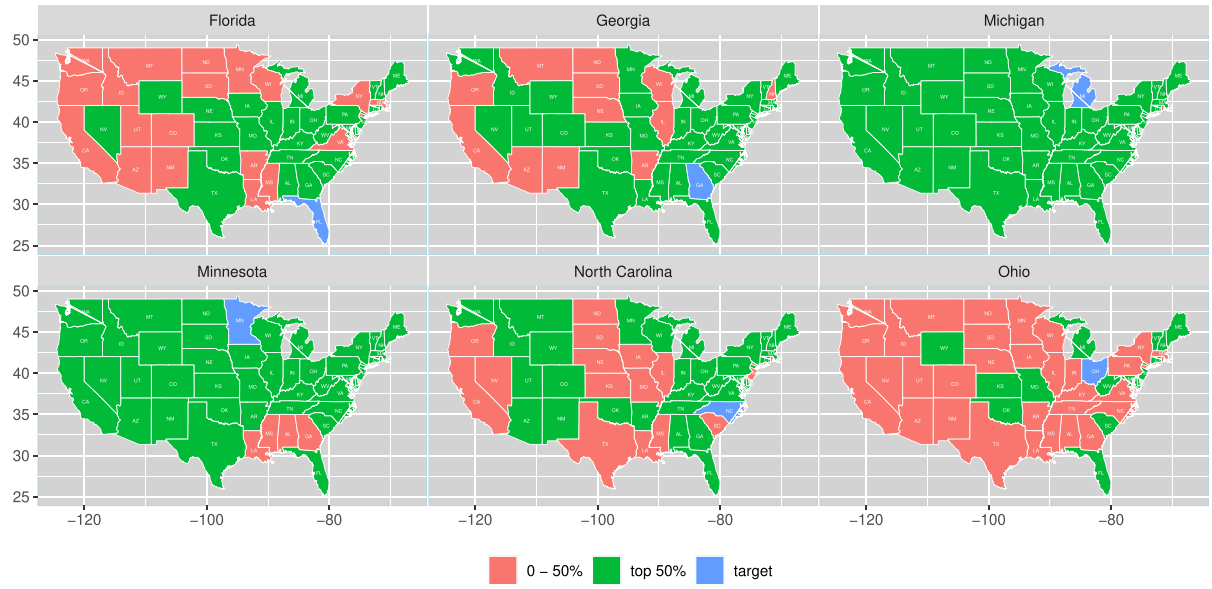


Figure 4. The detected transferable states for each swing state.

swing state, we record the frequency of occurrences for all transferable states and depict states with more than 50% occurrences on the map in green color. As observed in Figure 4, some transferable states exhibit geographical information signals. However, for some targets, the spatial or geographical correlations may not be readily apparent. This is a commonplace occurrence. With the development of the economy, culture, and scientific technology, regional interconnections are no longer confined solely to spatial proximity. Latent economic and political factors become pivotal.

In addition, to provide a comprehensive demonstration of the superiority of the proposed TranSAR method, we treat each state as the target once a time. Subsequently, we conduct both the tranSAR and SAR procedures, and the outcomes have been summarized in supplementary materials. In conclusion, our tranSAR method consistently outperforms the traditional SAR method in the majority of cases, yielding superior prediction results.

Motivated by public interest, we also predict the outcome of the 2024 U.S. presidential election for all states using demographic and geographical data in 2022, based on the trained tranSAR model from the 2020 U.S. voting data along with other demographic and geographical data. We use the 2022 demographic data and the voting data of the last election as covariates to predict county-level support rates. And we use the 2020 total vote counts of county-level as weights to calculate state-level support rates (because in the U.S., the "winner-takes-all" system weights each county by its vote count). We focus on the predictions of election outcomes for eight "swing states" following a rule based on whether there was a change in the supported party over the past *three* elections. Therefore, we select Arizona, Georgia, Florida, Pennsylvania, Michigan, Wisconsin, Ohio, and Iowa as swing states. For nonswing states, we use the results from their last three presidential elections as the outcome for this election, while for swing states, we use our tranSAR method for prediction. Considering the randomness inherent in the algorithm, we conduct multiple rounds of analysis using our TranSAR procedure. In these analyses, if the Democratic Party receives support in more than half of the iterations, the swing state was classified as supporting the Democratic Party; otherwise, it was classified as supporting the Republican Party. Iowa and Ohio are predicted to support the Democratic Party, while the other swing states are predicted to support the Republican Party. Among these swing states, except for Ohio, the swing states are predicted correctly compared to the 2024 election results in the real election. We obtain the supporting party for each state (see Figure 5), and calculate the final support vote counts combined with the electoral votes. The final prediction result shows that the Democratic Party receives 255 electoral votes, less than the threshold of 269 votes. Therefore, we predict that the outcome of the 2024 U.S. presidential election will favor the Republican Party.

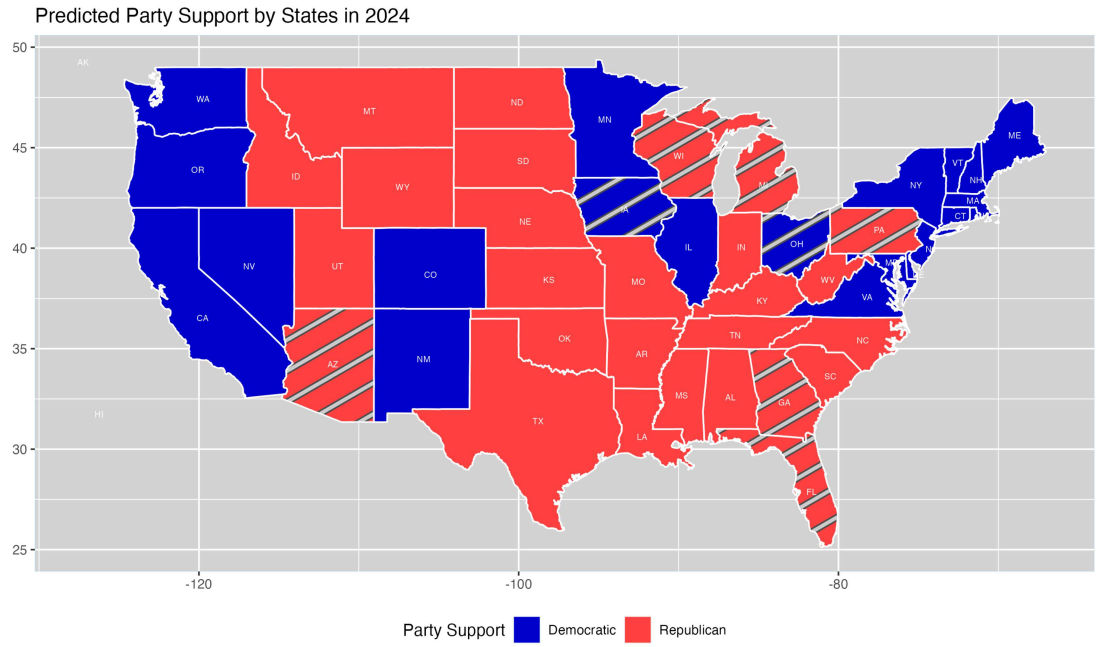


Figure 5. Predicted Party support by states in 2024. The striped states are the swing states whose election outcomes predicted by our TranSAR method.

6.1. Discussion on Limitations and Risks

While the proposed tranSAR method demonstrates superior performance in the analysis of the 2020 and 2024 election, it is crucial to interpret these results with caution. Election forecasting involves complex sociopolitical dynamics that may not be fully captured by statistical models. We discuss the potential deviations between forecasts and actual outcomes as follows:

6.1.1. Inherent Model Limitations

Our framework relies on a linear spatial autoregressive structure, facilitating theoretical and computational efficiency but potentially overlooking nonlinear voting behaviors. Specifically, socioeconomic impacts may exhibit heterogeneity or threshold effects rather than constant marginal effects. Furthermore, assuming a single spatial spillover parameter λ may oversimplify local variations in spatial dependence.

6.1.2. Impact of Model Misspecification

Forecast accuracy relies on the correct specification of confounding variables and the spatial weight matrix. First, the **omission of confounding variables**, such as unobserved local shocks or candidate charisma, may introduce endogeneity bias despite our inclusion of extensive covariates. Second, the **specification of spatial dependence structures** via geographic contiguity may overlook modern networks like social media, potentially biasing estimated spillover effects.

6.1.3. Impact of "Black Swan" Events

Finally, statistical models are vulnerable to "black swan" events, rare, unpredictable shocks like pandemics or political assassination, that disrupt data-generating processes. Since our framework assumes temporal stability, it cannot adjust for unprecedented anomalies. Thus, predictions serve as baseline inferences under a stable sociopolitical environment rather than absolute prophecies.

Despite these inherent risks, the tranSAR framework offers a powerful and innovative solution for election forecasting, especially with limited samples and complex spatial dependencies. It retains the interpretability of linear models, enabling explicit quantification of socio-economic and spatial effects. Furthermore, the framework is highly extensible to more complex spatial specifications. Thus, tranSAR represents a forward-looking contribution balancing methodological rigor with practical applicability.

7. Conclusion

To more accurately explore the spatial factors influencing swing states, we propose a method to enhance the estimation of the spatial model in swing states by transferring knowledge from other states. We introduce a transfer learning method within the framework of SAR models, aimed at addressing spatially dependent data of a small sample size. The empirical application of this methodology, specifically in predicting U.S. election results in swing states using demographic and geographical data, demonstrates its potential to enhance traditional spatial methods. Based on our comprehensive analysis and the application of the tranSAR model, our prediction for the 2024 U.S. presidential election is a victory for the **Republican Party**. Future research could focus on developing tests to detect informative sets and improving the accuracy of coefficient estimation within the transfer learning framework.

Supplementary Materials

All technical proofs of theorems and the additional simulations are included in the online supplementary materials. The code is available at https://anonymous.4open.science/r/tranSAR-E508.

Disclosure Statement

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